Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick the correct answer from the following.

1. Which one of the following statements is false about dynamic equilibrium?
   a. It takes place in a close container
   b. Concentration of reactant and products are not changed
   c. Rate of forward reaction is equal to rate of reverse reaction
   d. Equilibrium cannot be disturbed by any external stress

2. Consider the following reaction and indicate which of the following best describe equilibrium constant expression Kc.

$$4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_20$$

   a. $Kc = \frac{[NO]^4\,[H_2O]^6}{[NH_3]^4\,[O_2]^5}$
   b. $Kc = \frac{[NH_3]^4\,[O_2]^5}{[NO]^4[H_2O]^6}$
   c. $Kc = \frac{[NH_3][O_2]}{[H_2O][NO]}$
   d. $Kc = \frac{[4NO][6H_2O]}{[4NH_3][5S_2]}$

3. A reaction which is never goes to completion is known as reversible reaction. Reversible reaction is represented by:
   a. Doted lines
   b. Single arrow
   c. Double arrow
   d. Double straight line

4. When the magnitude of Kc is small, indicates
   a. Reaction mixture contain most of the reactant
   b. Reaction mixture contain most of the product
   c. Reaction mixture contain almost equal amount of reactant and product
   d. Reaction goes to completion

Sindh Textbook Board Jamshoro

5. For which system does the equilibrium constant, Kc has units of concentration

a. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

b. $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$

c. $H_{2(g)} + I_{2(g)} \rightleftharpoons 2\,HI_{(g)}$

d. $CO_{2(g)} + H_{2(g)} \rightleftharpoons CO_{(g)} + H_2O_{(l)}$

6. The unit of Kc for reaction $N_2 + O_2 \rightleftharpoons 2NO$

a. mol dm^{-3}

b. $mol^{-2}\ dm^6$

c. $mol^{-1}\ dm^3$

d. no unit

7. The system is stable in equilibrium when:

a. $Q_c = K_c$

b $Q_c > K$

c $Q_c < K$

d None of these

8. Qc can be defined as:

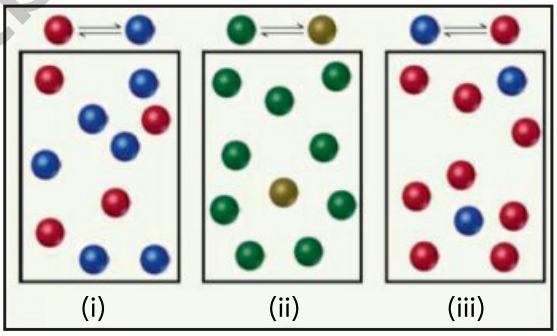
a. ratio of product and reactants

b. ratio of molar concentration of product and reactant at specific time

c. ratio of molar concentration of product and molar volume of reactant

d. ratio of molar concentration of product and reactant raised to the power of coefficient

9. Which of the following represent backward reaction?



a. (i) and (ii)

b. (ii) and

c. (ii) only

d. (iii) o nly

Sindh Textbook Board Jamshoro

10. The value of Kc increases when:
    a. [Product] is less
    b. [Product] is more
    c. [Reactant] is more
    d. [Reactant = product]

SECTION- B: SHORT QUESTIONS:

1. Define chemical equilibrium with example.
2. Why chemical equilibrium is dynamic?
3. When writing an equation, how is a reversible reaction distinguished from irreversible reaction?
4. Write an equilibrium equation of monoatomic carbon and a molecule of oxygen as reactant and carbon monoxide as product.
5. Outline the characteristics of reversible reaction.
6. Distinguished between reversible and irreversible reaction.
7. State law of mass action. How is the active mass is represented?
8. Why equilibrium constant may or may not have unit? Justify with example.
9. How direction of a reaction can be predicted if Kc is known to you.
10. Write equilibrium constant expression for the following equations:
    i) $N_2 + 2O_2 \rightleftharpoons 2NO_2$
    ii) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
    iii) $H_2 + Br_2 \rightleftharpoons 2HBr$

SECTION- C: DETAILED QUESTIONS:

1. Describe dynamic equilibrium with two examples.
2. State law of mass action. Derive an expression for equilibrium constant.
3. Describe the characteristics of equilibrium constant.
4. How can you predict the following stages of a reaction by comparing the values of Kc and Qc.
    i) Net reaction proceeds in forward direction.
    ii) Net reaction proceeds in reverse direction
5. Predict which system at equilibrium will contain maximum amount of product and which system will contain maximum amount of reactant?
    a) $2CO_{2(g)} \rightleftharpoons 2CO_{(g)} + O_{2(g)}$ $Kc(927^{\circ}C) = 3.1 \times 10^{-18}$ $mol.dm^{-3}$
    b) $2O_{3(g)} \rightleftharpoons 3O_{2(g)}$ $Kc(298K) = 5.9 \times 10^{55}$ $mol.dm^{-3}$

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

Encircle the correct answer in each case.

1. **Corrosive effect on skin is caused by:**

(a) acid (b) base (c) salt (d) both a & b

2. **Preservatives are used to preserve :**

(a) acids (b) bases (c) food (d) water

3. **Which of the following is not an Arrhenius acid:**

(a)HCl (b) CO_2 (c) HNO_3 (d) H_2SO_4

4. **NH_3 can be a base according to :**

(a) Arrhenius theory (b) Bronsted- lowery theory

(c) Lewis theory (d) both b and c

5. **Which of the following is a Lewis base?**

(a) HNO_3 (b) CN^- (c) HCl (d) $AlCl_3$

6. **A substance that can donate a pair of electron to form coordinate covalent bond:**

(a) Lewis acid (b) Lewis base

(c) Bronsted- Lowery acid (d) Bronsteed-Lowery base

7. **If pH value is greater than 7, then solution is :**

(a) acidic (b) basic (c) amphoteric (d) neutral

8. Salt among following is :

(a) HCl (b) KCl (c) HNO_3 (d) H_2SO_4

9. **Substances that react with both acids and bases are called :**

(a) conjugate acids (b) conjugate bases

(c) amphoteric substances (d) Buffers

10. **The reaction of acid and base to form salt and water is called :**

(a) hydration (b) Neutralization (c) hydrolysis (d) both a & c

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. Encircle the correct answer:

(i) The branch of chemistry which deals with the hydrocarbons and their derivatives is known as:

(a) Organic chemistry (b) Inorganic chemistry (c) Biochemistry (d) Physical chemistry

(ii) The general formula for alkanes is:

(a) C_nH_{2n} (b) C_nH_{2n+1} (c) C_nH_{2n+2} (d) C_nH_{2n-2}

(iii) Which of the following is an alcohol?

(a) CH_3CHO (b) $CH_3CH_2O\text{-}CH_3$ (c) CH_3OH (d) HCOOH

(iv) Which of the following is saturated hydrocarbon?

(a) $CH_3CH{=}CH_2$ (b) $CH_3CH_2CH_3$ (c) $CH_3C{\equiv}CH$ (d) $CH_2{=}CH\text{-}C{\equiv}CH$

(v) The prefix 'hept' stands for the …. carbon atoms.

(a) 2 (b) 5 (c) 7 (d) 9

(vi) The functional group –COOH is used for:

(a) Alkynes (b) alcohols (c) Phenols (d) Carboxylic acids

(vii) Polyethene is:

(a) oil (b) paper (c) plastic (d) wood

(viii) Acetic acid is obtained from:

(a) Banana (b) Dates (c) Garlic (d) Vinegar

(ix) Alkenes:

(a) show same general formula as alkynes.
(b) have carbon carbon triple bond.
(c) have carbon carbon double bond.
(d) are saturated hydrocarbons.

(x) $CH_3\text{-}CH_2\text{-}$ is … radical.

(a) Methyl (b) Ethyl (c) n-propyl (d) Isopropyl

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

(i) **Glucose is:**

(a) vitamin (b) protein (c) carbohydrate (d) lipid

(ii) **The deficiency of vitamin D causes:**

(a) beriberi (b) rickets (c) scurvy (d) haemorhage

(iii) **….. encodes a genetic information.**

(a) R.N.A (b) D.N.A (c) progesterone (d) cholesterol

(iv) **The carbohydrates which contain aldehyde group are called:**

(a) sacchrides (b) ketoses (c) pentoses (d) aldoses

(v) **Amino acids are building blocks of:**

(a) nucleic acids (b) protein (c) vitamins (d) lipid

(vi) **Which one of the following is polysacchride.**

(a) Fructose (b) maltose (c) starch (d) None of these

(vii) **Lactose is:**

(a) Grape sugar (b) honey sugar

(c) milk sugar (d) cane sugar

(viii) **What is true about a peptide?**

(a) It is a protein
(b) It is an anhydride of a carboxylic acids
(c) It is an anhydride of an amine
(d) It is a polyamide.

(ix) **Fats are solids at:**

(a) ordinary room temperature (b) high temperature

(c) higher than 50°C temperature (d) None of these

(x) **Cotton contains … cellulose.**

(a) 30% (b) 65% (c) 85% (d) 95%

SECTION- B: SHORT QUESTIONS:

(i) What are the proteins?

(ii) Define the importance of deoxyribose nucleic acid (DNA).

(iii) Differentiate between fats and oils.

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. Second highest layer of Earth's atmosphere is
   (a). stratosphere (b). mesosphere (c). troposphere (d). thermosphere
2. Aeroplanes fly in:
   (a). Troposphere (b). Stratosphere (c). Mesosphere (d). Thermosphere
3. Atmospheric pressure decreases with the
   (a). increase in longitude (b). decrease in altitude
   (c). increase in altitude (d). increase in latitude
4. Layer of atmosphere which separates stratosphere and troposphere is known as
   (a). tropo-pause (b). mesopause (c). thermopause (d). stratopause
5. Ozone layer is part of
   (a). mesosphere (b). stratosphere (c). thermosphere (d). troposphere
6. Which is not part of greenhouse gases
   (a). carbon dioxide (b). methane (c). nitrous oxide (d). oxygen
7. Second most abundant constituent of dry air in terms of volume after nitrogen is
   (a). nitrogen (b). oxygen (c). carbon dioxide (d). helium
8. Which of the following is the reason of global warming
   (a) Presence of sulphite
   (b) Rise in CO_2 concentration.
   (c) Oxides of nitrogen
   (d) Formtion of ozone
9. The altitude on stratosphere is:
   (a). 40 to 45 km (b). 50 to 55 km (c). 60 to 65 km (d). 70 to 75 km
10. Ozone is a gas found in the _____ layer:
    (a) Troposphere
    (b) Mesosphere
    (c) Stratosphere
    (d) Exosphere

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. Which of the following water borne diseases is of viral origin.
   (a) Typhoid fever (b) Polio (c) Dysentery (d) diarrhea
2. How much percentage (%) of the Earth's Surface is covered with water?
   (a)70% (b) 60% (c) 90% (d) 75%
3. Which type of bond is formed between H_2O molecules:
   (a) Hydrogen bond (b) ionic bond (c) covalent bond (d) all of these
4. The permanent hardness of water is due to presence of:
   (a) $MgSO_4$ (b) $Mg(HCO_3)_2$ (c) $Ca(HCO_3)_2$ (d) all of these
5. How much fresh water is present on earth:
   (a) 0.3% (b) 3% (c) 0.2% (d) 2%
6. Which salts are excessively dissolved to make temporary hard water:
   (a) $CaSO_4$ and $CaCl_2$ (b) KNO_3 and KOH
   (c) $CaCO_3$ and $Ca(OH)_2$ (d) $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$
7. Water is a:
   (a) Polar solvent (b) Non polar solvent
   (c) Amphipathic solvent (d) Non polar charged solvent
8. The taste of water is:
   (a) Sour (b) Bitter (c) Sweet (d) Tasteless
9. Which of the following is helpful for removal of permenant hardness:
   (a) Na_2CO_3 (b) $Ca(OH)_2$ (c) $CaCO_3$ (d) Na_2SO_4

SECTION- B: SHORT QUESTIONS:

1. Describe composition of water.
2. Define hard and soft water
3. Describe water pollutant.
4. How we can remove temporary hardness of water?
5. List down the diseases due to polluted drinking water.
6. Differentiate between hard and soft water.

SECTION- C: DETAILED QUESTIONS:

1. Write down the methods for removal of permanent hardness of water.
2. Describe the water pollutants in industries.
3. Justify that "Water is solvent".
4. Describe disadvantages of hard water?
5. Explain in detail Water borne diseases.

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. Encircle the correct answer:

1. The Analytical Chemistry deals with instruments and methods to _____, identify and quantify the matter.
   (a) MIx (b) Separate (c) Differentiate (d) Manipulate
2. The sample may be solid, liquid, gas or a _____ in qualitative analysis.
   (a) Mixture (b) Compound (c) Substance (d) None of these
3. Analysis deals with the identification of presence of functional groups in compounds is.
   (a) Physical qualitative analysis (b) Analytical qualitative analysis
   (c) Organic qualitative analysis (d) Inorganic qualitative analysis
4. Flame test of Copper Halide with bluish-green color identify the presence of.
   (a) Halogen (b) Hydrogen (c) Copper (d) b and c
5. The physical methods used to measure physical properties is called
   (a) Combustion analysis method (b) Atomic emission spectroscopy method
   (c) Volumetric analysis method (d) Gravimetric analysis method
6. The error caused by improper functioning of instrument is:
   (a) Determinant Error (b) In determinant Error
   (c) Systematic Error (d) Both a & c
7. An agreement between a measured value and the accepted true value.
   (a) Error (b) Accuracy (c) Precision (d) All of these
8. Spectroscopy is the interaction of light with :
   (a) Liquid (b) Solid (c) Gas (d) Matter
9. The Gas is mobile phase in:
   (a) Liquid chromatography (b) Solid chromatography
   (c) Gas chromatography (d) None of these
10. It used to assess concentration or amount of given atomic, molecular or ionic chemical.
   (a) Chromatography (b) Spectroscopy
   (c) Conductometry (d) Potentiometry

Sindh Textbook Board Jamshoro

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

1. Soap is the term for a salts of a:

(a) Carboxylic acid (b) citric acid (c) Sulphuric acid (d) fatty acid

2. Surfactants reduce the _____ of water.

(a) Viscosity (b) Surface tension (c) Boiling point (d) Melting Point

3. The carboxylate end of the soap molecule that is attracted to water is called __.

(a) hydrophobic end (b) end point (c) hydrophilic end (d) n.o.t

4. The use of potassium hydroxide produced a:

(a) Hard soap (b) Soft soap (c) Moderate soap (d) All of these

5. The citric acid is used in preparation of cold drinks for:

(a) Sweet taste (b) Bitter taste (c) Sour taste (d) Salty taste

6. The centrifuge machine used for separation of:

(a) Juice (b) pH (c)Mud (d)Crystal

7. The abrasives are:

(a) Water soluble minerals
(b) Water insoluble minerals
(c) Water semi soluble minerals
(d) Water absorving minerals

8. The harvesting is most important step of:

(a) Preparation of soap
(b) Preparation of coldrinks
(c) Preparation of sugar
(d) Preparation of medicines

9. Which of the following is used as jet fuel:

(a) Kerosene oil (b) Diesel oil (c) fuel oil (d) petrol

10. Which one of the following is not a fraction of crude oil?

(a) paraffin wax (b) ammonia (c) fuel oil (d) petroleum coke

Sindh Textbook Board Jamshoro